Homework 5: Concise Answers

A (p. 1): Proof in words

With the bar denoting negation, the statement is

$$[(a \vee b) \wedge (\neg a \vee c)] \rightarrow \neg(b \vee c).$$

The statement is false as written. Suppose a and c are true and b is false. The first clause is true because a is true, and the second clause is true because c is true. Thus the premise is true. However, $b \vee c$ is true, so its negation is false. This gives a true premise and a false conclusion, making the implication false. Therefore it cannot be proved as a valid implication.

$$\boxed{a = T, \quad b = F, \quad c = T \quad \text{is a counterexample.}}$$

The valid resolution conclusion would be $b \vee c$.

B (p. 1, second clause set): Resolution

Number the given clauses $C_1, \ldots, C_6$. Each subsequent row is the resolvent of the indicated two clauses.

Clause	Expression	Reason
C_1	$p \vee q \vee \neg r \vee s$	Given
C_2	$\neg p \vee r \vee s$	Given
C_3	$\neg q \vee \neg r$	Given
C_4	$p \vee \neg s$	Given
C_5	$\neg p \vee \neg r$	Given
C_6	r	Given
C_7	$\neg q$	C_3, C_6; resolve on r
C_8	$\neg p$	C_5, C_6; resolve on r
C_9	$\neg s$	C_4, C_8; resolve on p
C_{10}	$p \vee q \vee s$	C_1, C_6; resolve on r
C_{11}	$p \vee s$	C_{10}, C_7; resolve on q
C_{12}	s	C_{11}, C_8; resolve on p
C_{13}	$\square$ (empty clause)	C_{12}, C_9; resolve on s

The empty clause is false under every assignment. Since resolution is sound, the original clause set is $\boxed{\text{unsatisfiable}}$. The contradiction already follows from C_1, C_3, C_4, C_5, C_6; C_2 is not needed.

C (p. 4, Exercise 58): Simultaneously true disjunctions

Let

$$D_1 = p \vee \neg q, \quad D_2 = \neg p \vee q, \quad D_3 = q \vee r,$$
$$D_4 = q \vee \neg r, \quad D_5 = \neg q \vee \neg r.$$

Choose $p = T,\ q = T,\ r = F$. Then

$$D_1 = T \vee F = T, \quad D_2 = F \vee T = T,$$
$$D_3 = T \vee F = T, \quad D_4 = T \vee T = T,$$
$$D_5 = F \vee T = T.$$

All five are simultaneously true. Since there are only five disjunctions,

The maximum is 5.

D (p. 5): Equations and implications

Equations. Over the real numbers,

$$x = y, \quad y = z, \quad z = -x \quad \Longrightarrow \quad x = y = z = -x \quad \Longrightarrow \quad 2x = 0.$$

Hence $x = y = z = 0$. Conversely, this assignment satisfies all three equations. Thus yes, exactly when $x = y = z$

Propositional logic. Let

$$\Phi = (A \to B) \wedge (B \to C) \wedge (C \to \neg A).$$

An implication is false only when its antecedent is true and its consequent is false.

A	B	C	$A \to B$	$B \to C$	$C \to \neg A$	Φ
T	T	T	T	T	F	F
T	T	F	T	F	T	F
T	F	T	F	T	F	F
T	F	F	F	T	T	F
F	T	T	T	T	T	T
F	T	F	T	F	T	F
F	F	T	T	T	T	T
F	F	F	T	T	T	T

Choice (b): true simultaneously in some cases, but not all.

Exactly three assignments work: $(A, B, C) = (F, T, T), (F, F, T), (F, F, F)$. Equivalently, $\Phi \equiv \neg A \wedge (\neg B \vee C)$, which is satisfiable but is not a tautology.